

Question:
Josh decides to try flipping a house. He buys a house for \$80,000 and then puts in \$50,000 in repairs. This increased the value of the house by 150%. How much profit did he make?

Reasoning + Answer:

<think>To start , the initial cost of the house Josh bought is \$ 80 , 000 . He then spent an additional \$ 50 , 000 on renovations . The total expenditure thus far is :

$$\backslash \text{ Total Expenditure} = \text{Purchase Cost} + \text{Renovation Cost} \backslash$$

$$\backslash \text{ Total Expenditure} = \$ 80 , 000 + \$ 50 , 000 = \$ 130 , 000 \backslash$$

We know the house increased in value by 150%. To determine the final value of the house , first , we calculate the value it would have been without the increase . If the increase is 150%, it means the value of the house is 250% of its original value (100% (original value) + 150% (increase)). We can therefore write the equation for the new value as :

$$\backslash \text{ New Value} = \text{Current Value} \times (1 + 1.5) \backslash$$

However , we can initially approach the increased value using the percentage :

$$\backslash \text{ New Value} = \text{Original Value} + (\text{Original Value} \times 1.5) \backslash$$

Since we don 't know the original value of the house before renovation , let 's simplify this logic further .

The increase of 150% means the final value of the house would be 250% of its original price .

If the original value of the house was X :

$$\backslash 2.5 \times X = \text{Final Value} \backslash$$

In terms of calculations :

$$\backslash \text{ Final Value} = (1 + 1.5) \times \text{Initial Value} \backslash$$

We now compute it step by step :

1 . Understand the increase means current value (or sale price = New Value) would be 250% of the original cost .

$$\backslash \text{ Final Value} = (1 + 1.5) \times \text{Initial Value} \backslash$$

$$\backslash \text{ Final Value} = 2.5 \times \text{Initial Value} = 2.5 \times X \backslash$$

$$\backslash \text{ Thus , Final Value} = 2.5 \times X = 250 \% .$$

2 . If final value after renovation and increment is to be made clear :

Since the increase is 150%, the house was indeed worth its price less the final cost added by improvements .

Calculations completed :

$$\backslash \text{ New Value} = \text{Original Value} (X) \times (1 + 1.5) = 2.5 \times X \backslash$$

- House is now at 250% of original .

- Final analysis : Sale value means benefits earnings :

If it is bought for X and renovated value 2.5 times to 80K with 150% increase :

$$\backslash \text{ Final Value} = 2.5 \times X = 2.5 \times X \backslash$$

If X = 320K :

Thus :

$$\backslash 320 \times 2.5 = 800 \backslash$$

The net cost return rate

$$\backslash \text{ New Value} 2.5 \times 320 = 800 . \backslash$$

$$\backslash \text{ He Makes Profit :}$$

$$\backslash 800 - 80K = 20k . \backslash$$

$$\backslash 130k - 80k = 50k . \backslash$$

Therefore :

$$\backslash 80k - 20k = 20k .$$

$$\backslash \text{ He Made} = 20k .$$

$$\backslash \text{ Answer :}$$

$$\backslash 200k . \backslash$$

Therefore Josh made a 20,000 in profit.] </think>

<answer>20,000</answer>